



6120 Executive Blvd, Suite 550,
Rockville, MD 20852
Phone: 240 880-2004
e-mail: sbrown@genetics-gsa.org
www.genetics-gsa.org

March 6, 2026

Adam Cheong, BS or BA
Proof School
240 Exeter Ave
San Carlos, CA 94070
United States

Dear Adam Cheong,

On behalf of the Organizers, you are cordially invited to participate in the Population, Evolutionary, and Quantitative Genetics Conference to be held June 9-12, 2026 at Asilomar Conference Grounds in Pacific Grove, CA. The organizers greatly appreciate the important contribution that you will make to the meeting. The organizers greatly appreciate the important contribution that you will make to the meeting. All meeting attendees are required to pay a meeting registration fee and are responsible for all travel costs. The total cost of registration, housing and meals for the conference depends on the type of accommodations reserved and your registration category.

Your abstract entitled "How Cancer Cells Undergoing a Treatment Course Evolve to Survive Multiple Mass Extinctions" has been accepted and will be programmed. The Genetics Society of America is a professional membership organization for an international community of biologists advancing the field of genetics. The purposes of the Society are 1) to facilitate communication between geneticists, 2) to promote research that will bring new discoveries in genetics, 3) to foster the training of the next generation of geneticists so they can effectively respond to the opportunities provided by our discoveries and the challenges posed by them, and 4) to educate the public and their government representatives about advances in genetics and the consequences to individuals and to society.

The National Academies has a useful web site concerning obtaining a visa (<http://national-academies.org/visas>) for international participants. As part of new security procedures, many applications are being sent to the State Department in Washington where they are reviewed, with assistance from other agencies. Because of the number of visas being processed and the need to be thorough with the reviews, this can take as much as 3 to 4 months or more. Therefore, we advise scientists intending to come to the United States to apply for their visa as early as possible.

Sincerely,

Suzy Brown
Senior Director of Conferences
6120 Executive Blvd, Suite 550, Rockville, MD 20852
Phone: 240 880-2004
e-mail: sbrown@genetics-gsa.org
www.genetics-gsa.org

How Cancer Cells Undergoing a Treatment Course Evolve to Survive Multiple Mass Extinctions

The vast diversity of genetic and phenotypic characteristics observed in human cancers, according to Hanahan (2026), can be understood in the context of how cancers arise via multistep pathways of tumorigenesis that lead to the formation of primary tumors, which subsequently evolve and progress due to selective advantage in the face of multifaceted barriers to continuing proliferative expansion, resulting in metastasis and adaptive resistance.

Here we elucidate the adversarial viewpoint: a press-pulse framework borrowed from paleobiology (Arens & West, 2008) for modeling therapeutic intervention targeted at the primary tumor, whose microenvironment undergoes terraforming and subsequently becomes unfit for cancer cells proliferation, leading to multiple mass extinctions over the course of cancer treatment, finally resulting in remission as the residual cancer cells turn quiescent.

The population dynamics driven by tumor growth vs. dose response is captured in our proposed model, as the cancer cells population follows a Gompertzian growth curve after metabolic reprogramming but shrinks in response to drug treatment. In particular, to shrink a tumor over multiple treatment cycles, dose responses are modeled as the outcome of catastrophic events decimating cancer cells as intravenous antibody-drug conjugate (Madhusoodanan, 2024) diffuses across blood vessel walls and penetrates the tumor, driven by concentration gradient according to Fick's laws.

Tumoral evolution within our framework offers mechanistic insights into the structure of a particular fitness landscape characterized by the selective pressure upon residual regrowth. We illustrate how subpopulations of residual cancer cells evolve to survive multiple mass extinctions originated from a prescribed course of antibody-drug conjugate targeting the primary tumor, potentially laying the seeds for future metastasis or adaptive resistance. We propose to construct a simulation testbed to further explore cancer population dynamics to better understand the mechanism of relapse.

References:

1. Hanahan, D. (2026). Hallmarks of cancer—Then and now, and beyond. *Cell*, 189.
2. Arens, N. C., & West, I. D. (2008). Press-pulse: a general theory of mass extinction? *Paleobiology*, 34(4), 456–471.
3. Madhusoodanan, J. (2024, March 1). These cancers were beyond treatment—but might not be anymore. *Scientific American*.

How Cancer Cells Undergoing a Treatment Course Evolve to Survive Multiple Mass Extinctions

Adam Cheong^{1,2}, Mark Cheong², and Paul Fong³

¹Proof School, ²UC Berkeley, ³Duke University

Project Summary

The vast diversity of genetic and phenotypic characteristics observed in human cancers, according to Hanahan (2026), can be understood in the context of how cancers arise via multiplex pathways of tumorigenesis that lead to the formation of primary tumors, which subsequently evolve and progress due to selective advantage in the face of multifaceted barriers to continuing proliferative expansion, resulting in metastasis and adaptive resistance.

Here we elucidate the adversarial viewpoint: a *press-pulse* framework borrowed from paleobiology (Arens & West, 2008) for modeling therapeutic intervention targeted at the primary tumor, whose microenvironment undergoes *terroforming* and subsequently becomes unfit for cancer cells proliferation, leading to multiple mass extinctions over the course of cancer treatment, finally resulting in remission as the residual cancer cells turn quiescent.

The population dynamics driven by tumor growth vs. dose response is captured in our proposed model, as the cancer cells population follows a Gompertzian growth curve after metabolic reprogramming but shrinks in response to drug treatment. In particular, to shrink a tumor over multiple treatment cycles, dose responses are modeled as the outcome of catastrophic events decimating cancer cells as intravenous antibody-drug conjugate (Madhusoodanan, 2024) diffuses across blood vessel walls and penetrates the tumor, driven by concentration gradient according to Fick's laws.

Tumoral evolution within our framework offers mechanistic insights into the structure of a particular fitness landscape characterized by the selective pressure upon residual regrowth. We illustrate how subpopulations of residual cancer cells evolve to survive multiple mass extinctions originated from a prescribed course of antibody-drug conjugate targeting the primary tumor, potentially laying the seeds for future metastasis or adaptive resistance. We propose to construct a simulation testbed to further explore cancer population dynamics to better understand the mechanism of relapse.

Introduction

Antibody-Drug Conjugates Show Superior Response when Dosed at MTD

Over the last two decades, an increasing number of antibody-drug conjugates (ADCs) have been approved for cancer treatment, broadening therapeutic options for patients beyond traditional small molecule drugs. Through their unique mode of action as “magic bullets,” these targeted therapies hold the promise of redressing the poor balance between *safety and efficacy* with traditional chemotherapy^[A, R].

The “sweet spot” between the lowest concentration that produces a desired therapeutic effect (minimum effective dose or MED) and the concentration just before toxicity kicks in (maximum tolerated dose or MTD) is called the therapeutic window.

When dosed at or near MTD, ADCs display improved efficacy over small molecules in trials. Across the board, ADCs have already achieved greater beneficial outcomes for patients compared with traditional chemotherapy. The objective response rates (ORR) of ADC treatments frequently score above 40 percent in breast, gastric, urothelial, and lung cancer trials where small molecules rarely exceed 25 percent^[A, R].

Research Question

How to Make a Tumor Microenvironment “Most Unfit” Under Press-Pulse?

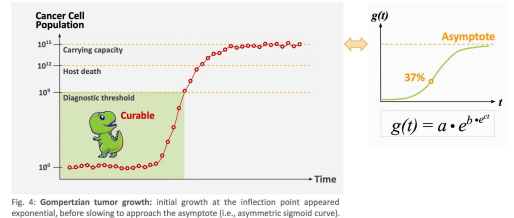
Four forces that drive cancer cells population dynamics:

- (a) **Tumor Growth:** proliferative cancer cells harness aerobic glycolysis to generate energy rapidly to drive tumor growth;
- (b) **Pulse Therapy:** dose responses are the outcome of catastrophic events decimating cancer cells as intravenous drug diffusing across blood vessel walls penetrates the tumor;
- (c) **Somatic Mutation:** somatic mutation during residual regrowth in between treatment cycles selects for drug-resistant cells that drive future cancer recurrence; and
- (d) **Press Therapy:** chronic microenvironmental stress (e.g., energy deprivation from starvation of glucose) slows down cell cycles and reduces cancer cells population resistance.

Note: Cancer cells compete for resources in order to grow rapidly, while evading counterattacks. Mathematical models of tumoral evolution describe how cancer cell populations grow with aerobic glycolysis but shrink under drug treatment, just as Lotka-Volterra equations describe the population dynamics of snowshoe hares and the lynx that feed upon them. New complexity in population dynamics arises when somatic mutation across treatment cycles and metabolic intervention are introduced and added to the model.

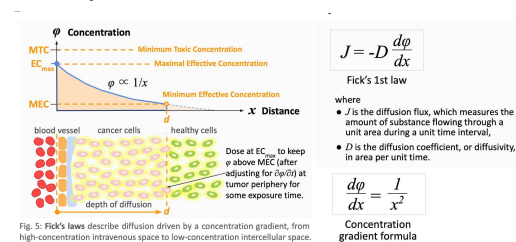
Methods — Growth

Modeling Tumor Growth with a Gompertzian Curve

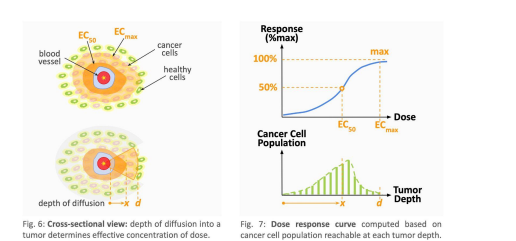


Methods — Pulse

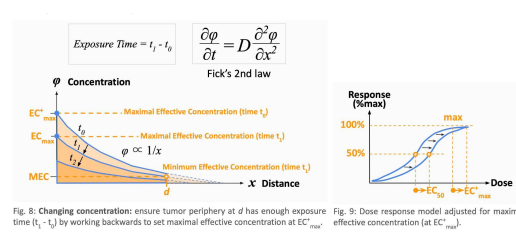
1 Dose at Maximal Effective Concentration — Dependent on Size of Tumor



2 Compute Dose Response Curve — for Each Tumor Size and Shape

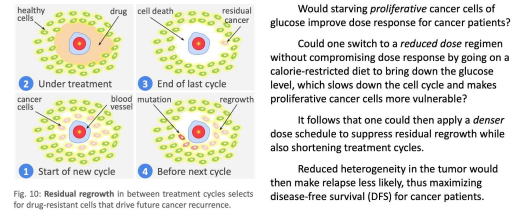


3 Adjust Dose Response Model — as Concentration Decreases Over Time



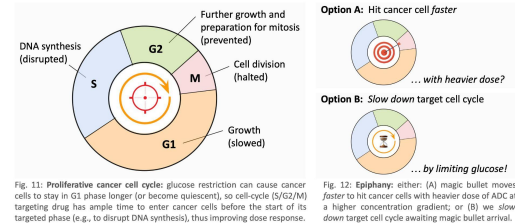
Discussion — Mutation

Cancer Cells Evolve in Between Treatment Cycles to Survive Mass Extinctions



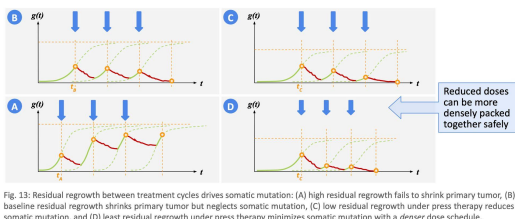
Discussion — Press

Making Proliferative Cancer Cells Vulnerable Through Bioenergetic Intervention



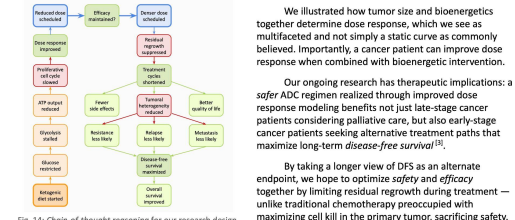
Conclusions

The Dose Makes the Poison: “Pulse Train” Designed to Suppress Regrowth



Future Work

Safer ADC Regimens That Also Maximize Disease-Free Survival?



References

- Arens, N. C., & West, I. D. (2008). Press-pulse: a general theory of mass extinction? *Paleobiology*, 34(4), 456–471.
- Aykan, N. F., & Ozati, T. (2020). Objective response rate assessment in oncology: Current situation and future expectations. *World Journal of Clinical Oncology*, 11(2), 53–73.
- Delgado, A., & Guddati, A. (2021). Clinical endpoints in oncology — a primer. *Am J Cancer Res*, 11(4), 1121–1131.
- Drago, J., & Norton, L. (2021, September). Implication of Cancer Growth Kinetics for the Further Development of Antibody-Drug Conjugates. *DIA Global Forum*.
- Hanahan, D. (2026). Hallmarks of cancer—Then and now, and beyond. *Cell*, 189.
- Madhusoodanan, J. (2024, March 1). These cancers were beyond treatment—but might not be anymore. *Scientific American*.
- Michor, F., & Beal, K. (2015). Improving Cancer Treatment via Mathematical Modeling: Surmounting the Challenges Is Worth the Effort. *Cell*, 163(5), 1059–1063.
- Mocny, J., Dieterich, P., & Glasford, I. (2023, July 4). Antibody Drug Conjugates: windows of opportunity. *Drug Target Review*.
- Scillia, K. R., Gallagher, K., Masud, M. A., Robertson-Tessi, M., Gatenbee, C. D., West, J., Llamas, P., Prabhakaran, S., Gallaher, J., & Anderson, A. R. A. (2025). Mathematical Oncology: How Modeling is Transforming Clinical Decision-Making. *Cancer Research*, 85(24), 4866–4879.
- Seyfried, T. N., Yu, G., Maroon, J. C., & DiGiovino, D. P. (2017). Press-pulse: a novel therapeutic strategy for the metabolic management of cancer. *Nutrition & Metabolism*, 14(1).
- Vasan, N., Baselig, J., & Hyman, D. M. (2019). A view on drug resistance in cancer. *Nature*, 575(7782), 299–309.